

	Title	Revision n°	date	Document No
	DETERMINATION OF SALICYLIC ACID (BETA HYDROXY ACID) IN COSMETIC PRODUCTS BY HIGH PERFORMANCE LIQUID CHROMATOGRAPHY (HPLC)	01	08/05/2025	009

1. SCOPE AND FIELD OF APPLICATION

This test method describes the determination of salicylic acid in cosmetic products (cream).

2. PRINCIPLE

Salicylic Acid is a Beta Hydroxy Acid (BHA). Salicylic acid is used in skin-care products for skin exfoliate and acne care. It is soluble in methanol which can be separated by HPLC. The concentration of salicylic acid in cosmetic products can be determined from its peak area in sample with the calibration curve of salicylic acid standard.

3. REAGENTS

3.1 General: all reagents used shall be of analytical purity.

- 3.1.1 Salicylic acid, Reference standard
- 3.1.2 Glacial acetic acid, AR grade
- 3.1.3 Methanol, HPLC grade (MeOH)
- 3.1.4 Water shall be distilled water (H₂O)

3.2 Standard stock solution

Prepare 0.1 % (w/v) solution of salicylic acid in methanol.

3.3 Standard calibration solutions

Prepare standard calibration solutions concentration of 2, 6, 10, 20 and 30 µg/mL in mobile phase. Pipette the standard solution 20, 60, 100, 200 and 300 µL, respectively with micropipette to each 10 mL volumetric flask. Make up to volume with mobile phase and mix well. Label as S1, S2, S3, S4 and S5, respectively.

3.4 Mobile phase: MeOH : 1.5% v/v glacial acetic acid (55 : 45 by volume)

Prepare 1000 mL of mobile phase, mix thoroughly 550 mL of methanol and 450 mL of 1.5% v/v glacial acetic acid and filter through 0.45 µm membrane filter (hydrophilic PVDF/PTFE). Degas for 5-10 mins.

4. APPARATUS

Normal laboratory equipment and:

- 4.1 High Performance Liquid Chromatograph (HPLC) with a photodiode array detector
- 4.2 Analytical column: Hypersil GOLD™ C18, 5 µm, 4.6 mm x 150 mm or equivalent.
- 4.3 Membrane filter, 47 mm Ø, 0.45 µm (PVDF or equivalent)
- 4.4 Disposable syringe filter 0.45 µm (PVDF or equivalent)
- 4.5 Ultrasonic bath
- 4.6 Electronic balance, 0.1 mg (readability)
- 4.7 Water bath capable of maintaining 60 °C
- 4.8 Vortex

5. PROCEDURE

5.1 Sample solution preparation:

- 5.1.1 Weigh accurately 0.25 g of sample (duplicate A and B) into 25 mL volumetric flask.
- 5.1.2 Dissolve with 10 mL of methanol by vortexing or sonicate or warm at 60 °C for 2-5 mins as necessary until cream is dispersed.
- 5.1.3 Let the sample cool down to room temperature, then make up to volume with mobile phase and mix well. Dilute quantitatively and stepwise, if necessary, with mobile phase.
- 5.1.4 Filter sample solution through disposable syringe filter (4.4) into vial with cap.

Note: The amount of the final concentration of sample solution shall be within calibration curve. If necessary, appropriate dilutions may be done as following guide.

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5.2 Preparation of spiked sample solution for determination of percentage recovery

5.2.1 Accurately weigh 0.25 g of sample (duplicate C and D) into 25 mL volumetric flask.

5.2.2 Add known amount of standard to the sample, then follow procedure 5.1.2 to 5.1.4.

Note: The amount of standard addition required matrix spike recoveries to fall 50-150% and the final concentration shall be within calibration curve.

5.3 HPLC conditions

5.3.1 Mobile phase: MeOH : 1.5% glacial acetic acid (55 : 45) by volume

5.3.2 Flow rate: 1.0 mL/minute

5.3.3 Photodiode array detection wavelength: 200 – 350 nm and $\lambda_{\max}$ at 236 or 302 nm

5.3.4 Column: Hypersil Gold™ C18, 5 μ m, 150 mm. x 4.6 mm. i.d

5.3.5 Injection volume: 20 μ L

5.3.5 Run time: 5 mins (for Standard calibration solutions) 8-15 mins (for sample solutions)

5.4 Sequence of injection to the HPLC system.

Sequentially inject the prepared solution to HPLC and record peak area as follows:

5.4.1 System suitability: Inject standard solution, S1 (3.3) to examine the retention time and triplicate injections to determine standard deviation (%RSD) of peak area and tailing factor. The acceptance criteria are as follows

Name	%RSD of peak area (n=3)	tailing factor
Salicylic acid	≤ 3	≤ 2.0

5.4.2 Inject S1, S2, S3, S4 and S5 (3.3), respectively for construction of the calibration curve.

5.4.3 Inject S3 (to compare the peak area with the peak area from calibration curve, % RPD should be < 3 %)

5.4.4 Inject mobile phase

5.4.5 Inject sample solution 1A, 1B, 1C, 1D

5.4.6 Inject S3 (to compare the peak area with the peak area from calibration curve, % RPD should be < 3 %)

5.4.7 Inject mobile phase

5.4.8 If there are more samples, inject S3 and mobile phase for each 4-10 injections. % RPD should be < 3 %)

5.4.9 Last vial is S3 to compare the peak area with the peak area from calibration curve, % RPD should be < 3 %)

6. CALCULATION

6.1 Construction of calibration curve between concentration and peak area of standard salicylic acid solutions. From linear regression equation:

$$A_0 = b_1 C_0 + b_0$$

Where

b_1 = slope

b_0 = intercept

C_0 = concentration of salicylic acid (μ g/mL)

A_0 = peak area

6.2 Calculation salicylic acid; SA

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6.2.1 Concentration of SA in sample solution

$$\text{SA, } \mu\text{g/mL} = [\text{Peak Area of sample} - \text{intercept, } b_0] / [\text{slope, } b_1]$$

6.2.2 Concentration of SA in sample in percentage by mass, using the formula:

$$\text{SA (\%w/w)} = [\text{SA, } \mu\text{g/mL} \times \text{dilution factor} \times 100] / [\text{sample weight (g)} \times 1 \times 10^6 \text{ ug/g}]$$

$$\text{dilution factor} = \frac{\text{initial volume of sample soln (ml)} \times \text{dilution volume (mL)}}{\text{volume from stock sample solution (mL)}}$$

6.3 % Recovery

$$\% \text{ recovery} = \frac{S - U}{C_{SA}} \times 100$$

Where,

S = concentration of salicylic acid in spiked sample, % w/w

U = concentration of salicylic acid in un-spiked sample, % w/w

C_{SA} = concentration of salicylic acid added, % w/w

6.4 % Relative Percent Difference

$$\% \text{ RPD} = \frac{\frac{A_2 - A_1}{A_1 + A_2}}{2} \times 100$$

Where,

A1 = peak area of S3 in calibration curve

A2 = peak area of S3 when inject interval in sequence of injection

7. REMARKS

7.1 Method validation information

7.1.1 Precision

7.1.1.1 Within day

Parameter	Cream product (%)
Within day (7 replicates): %RSD	0.04

7.1.1.2 Different days (Intermediate precision)

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Parameter	Cream product
Between-day (5 days/7 replicates each) : p-value	0.47

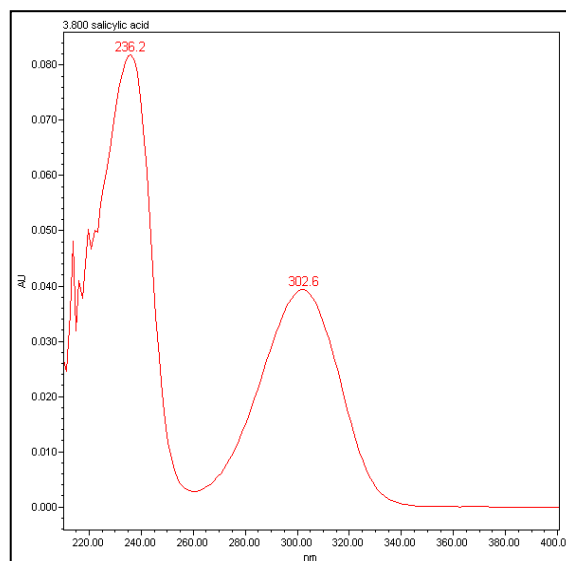
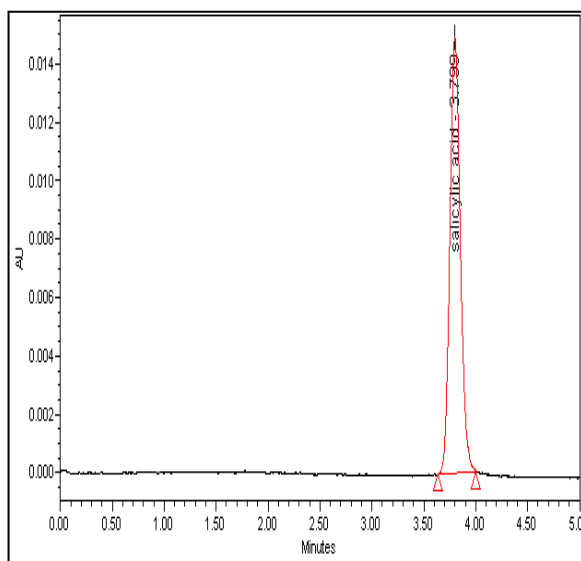
7.1.2 Limits

Parameter	Cream product (%w/w)
Limit of Detection (LOD)	0.005
Limit of Quantitation (LOQ)	0.01

7.1.3 Expanded uncertainty

Salicylic acid in	Concentration (%w/w)	Expanded uncertainty at 95% confidence level	
		% w/w	Relative
Cream product	5.00	0.45	0.04

7.2 Chromatogram and spectrum of BHA (salicylic acid)



Harmonised method:

Revised and Approved by AMS delegates on the 24th ACTLC Meeting, 28-29 April 2025.

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AMENDMENT RECORDS :

Revision no.	Amendment Detail	Revised by
01	- Section 5.4.1 - Removal of K-prime from the acceptance criteria as a mandatory parameter, making it optional. - Replacement of “methyl alcohol” with “methanol” throughout the document. - Editorial revisions throughout the document.	Thailand and AMS on the 24th ACTLC Meeting on 28-29 April 2025. Endorsed by the 41st ACC Meeting on 8 May 2025